

Hyperloop Awakening Whitepaper

A Non-Liquidative Lending Protocol with Option Attribution

Hyperloop Finance Research

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Abstract

Hyperloop Awakening is a non-liquidative, fixed-rate, fixed-maturity collateralized lending protocol implemented for the Ethereum Virtual Machine. It is organized around isolated, immutable, permissionlessly created markets, each parameterized by a collateral asset, a loan asset, a maturity date, and a strike. Lending and borrowing are implemented through the trading of credit and debt units whose payoff structure is analogous to that of zero-coupon obligations, settling at the market's maturity. The defining property of the protocol is that borrowers are never subject to interim liquidation: lender principal is protected by a put option attributed to each loan at origination, whose strike, notional, and expiry are aligned one-to-one with the market parameters. Liquidity is sourced through an offer-based market in the style of Morpho Midnight: makers publish executable offers that do not lock capital, and source the loan asset together with the corresponding option position only at settlement. The resulting design lets sophisticated makers concurrently quote across multiple maturities and strikes while hedging their exposure on existing on-chain and centralized option venues, and lets borrowers obtain fixed-term, fixed-rate financing without exposure to forced unwinds during the term of the loan.

1 Introduction

Crypto-collateralized lending has converged onto a small set of architectural choices: variable-rate, perpetual-term, oracle-driven, and liquidation-based [1, 2, 3]. These choices serve the use cases of short-term, leverage-seeking, liquid-collateral borrowers well, but exclude two categories of demand that today are served almost exclusively off-chain.

The first is the *long-duration holder*: a treasury, family office, or listed-company shareholder who holds a concentrated position in a digital asset, requires liquidity against it, and is unwilling to accept the path-dependent risk of being liquidated during a drawdown that the borrower considers temporary. For this borrower, the *expected* cost of liquidation -conditional on a 50% intra-loan drawdown that recovers before maturity- can exceed the entire interest cost of the loan. The economically rational response is to refuse the product, and the empirical response, historically, has been to seek bilateral OTC financing with structured downside protection.

The second is the *predictable-funding borrower*: a participant who needs to commit to a fixed cost of capital for a known period, typically because the use of proceeds itself has a fixed horizon. Variable-rate, perpetual-term pools cannot serve this need without imposing rate risk that defeats the use case.

Morpho Midnight [4] introduced an architecture for on-chain credit that addresses the second category: isolated, immutable, fixed-rate, fixed-maturity markets, with lending and borrowing implemented as the trading of fungible credit and debt units that settle at maturity. Makers post executable offers without locking capital, and source liquidity only at settlement, allowing a single maker to quote across many markets simultaneously.

Awakening extends this architecture to address the first category. We retain the Midnight market shape -isolated, immutable, fixed-rate, fixed-maturity, offer-based- and add a single structural property: every loan opened in a market is implicitly backed by a put option on the collateral, with strike, notional, and expiry aligned to the market's parameters. The put is the protocol's substitute for liquidation. A borrower is never subject to a margin call or forced unwind during the term of the loan; instead, if the collateral terminates below strike at maturity, the put pays out, the lender receives the full notional of their credit, and the borrower has the choice of repaying and recovering the collateral or walking away.

This single change has substantial consequences:

- The fixed loan term and the fixed option expiry coincide structurally. A 90-day loan against a 90-day put is one product, not two.
- The fixed loan rate naturally incorporates the put premium. Borrowers see one rate; makers price the put internally.
- The advance rate (LTV) is no longer a function of expected volatility multiplied by a liquidation buffer. It is a function of the strike chosen, full stop. Strikes can be set at 100% of spot or higher, because there is no liquidation event to defend against.
- The maker's quote is no longer constrained by their willingness to hold uncovered downside risk. Sophisticated makers can hedge the option exposure on external venues -Deribit, Derive, IBIT-listed contracts, or bilateral OTC- and quote against the *net* funding cost.

The remainder of this paper specifies the protocol. Section 2 defines the market primitive. Section 3 specifies the offer-based market layer. Section 4 details the option attribution mechanism. Section 5 specifies settlement at maturity. Sections 7 through 10 describe the supporting access-control,

authorization, and fee primitives, largely inherited from Midnight.

2 Fixed-maturity non-liquidative markets

2.1 Markets

Awakening is organized around isolated and immutable markets with configurations that cannot be altered after creation. Each market is uniquely specified by the tuple

$$M = \langle \text{collateral}, \text{loan}, \text{oracle}, \text{maturity}, \text{strike}, \text{settlement}, \text{gates} \rangle$$

where:

- *collateral* is the ERC-20 token posted by borrowers.
- *loan* is the ERC-20 token in which credit, debt, and interest are denominated.
- *oracle* is the price feed used at maturity for the strike comparison.
- *maturity* T_M is the timestamp at which the market settles. By convention T_M is chosen to coincide with a recognized option expiry calendar (weekly, monthly, or quarterly) on a reference option venue.
- *strike* K is the per-unit collateral price below which the embedded put position pays out at maturity. K is denominated in *loan* per *collateral*.
- *settlement* is the contract that executes the per-loan put attribution and maturity settlement (see Section 4).
- *gates* are optional access-control contracts (see Section 7).

A market's *advance rate* is implied by its strike: for a borrower posting one unit of collateral, the maximum debt that can be opened is bounded by

$$\text{maxDebt} = K$$

This identity is exact, and does not depend on an expected volatility input. It holds for any K ; markets may be created with K above the prevailing spot, allowing advance rates above 100%, provided makers are willing to quote against them.

2.2 Units

Within a market, positions are accounted in units. One debt unit is an obligation to repay one *loan* token at maturity. One credit unit is a claim on those repaid tokens, augmented by the protocol's settlement of the embedded put position. Buying credit units increases lender exposure; selling credit units decreases it. Symmetrically, borrowers sell debt units to increase their borrowing and buy them back to reduce it.

The implied user rate is read off the discount at which units are traded: for any traded price $P > 0$, the simple rate over the remaining term is

$$r = \frac{1}{P} - 1$$

This is identical to the unit semantics in Midnight, and is preserved deliberately, because the same makers are expected to operate across both protocols.

2.3 Fungibility

Every trade involves a buyer and a seller, but settlement results in fungible positions within the market rather than in a lasting bilateral relationship. Credit and debt are accounted for at the market level, and positions are not tied to the particular trade that created them.

As markets mature on fixed calendar dates rather than on rolling tenors from origination, positions created at different times but with the same T_M and K belong to the same market and are fungible with one another. The put attribution attached to each credit unit (Section 4) is also fungible at the market level: the lender’s protection is an aggregate of the market’s hedge inventory, not a bilateral position against any single borrower.

2.4 Early exits

Within a market, existing lenders and borrowers can reduce their outstanding credit and debt at any time by selling and buying units respectively. A buyer closes out any debt before accumulating credit, and a seller closes out any credit before accumulating debt. Early exits make users’ payoffs more flexible and deepen liquidity for all participants, since entries and exits occur within the same unified market.

Trading is notably possible after maturity, except that one cannot increase their debt then. This is enabled to facilitate unwinding in the event of contested settlement outcomes (Section 5.4).

3 Offer-based markets

Figure 1 summarizes the actors and asset flows of the protocol. The Awakening market is the central on-chain contract. Borrowers and lenders interact with the market directly. The maker is an off-protocol participant that mediates between the market and one or more external option venues: it sources put (and, in the generalized product of Section 11, call) positions externally and attributes them to the market at fill time. The oracle is a passive component, read only at maturity.

3.1 Offers

Makers use *offers* to specify the price and maximum size they are willing to trade on a given market. Offers are not broadcasted at the protocol level and may be distributed through any external channel, either offchain or onchain.

A taker executes an offer by submitting it to the Awakening contract. Takes may be partial: any size up to the offer’s remaining capacity is permitted, and a single offer can be filled by multiple takers until exhausted. The contract settles atomically against the referenced market, creating, transferring, or burning the corresponding credit and debt units, *and triggering the option attribution callback* (Section 4).

Each offer is attached with a *ratifier* contract that embeds the validation logic. It is called when an offer is taken. Typically, the ratifier verifies the validity of a signature of the offer against the maker’s public key. The modularity lets makers use different signature schemes such as passkeys or post-quantum schemes, or create more custom validation logic.

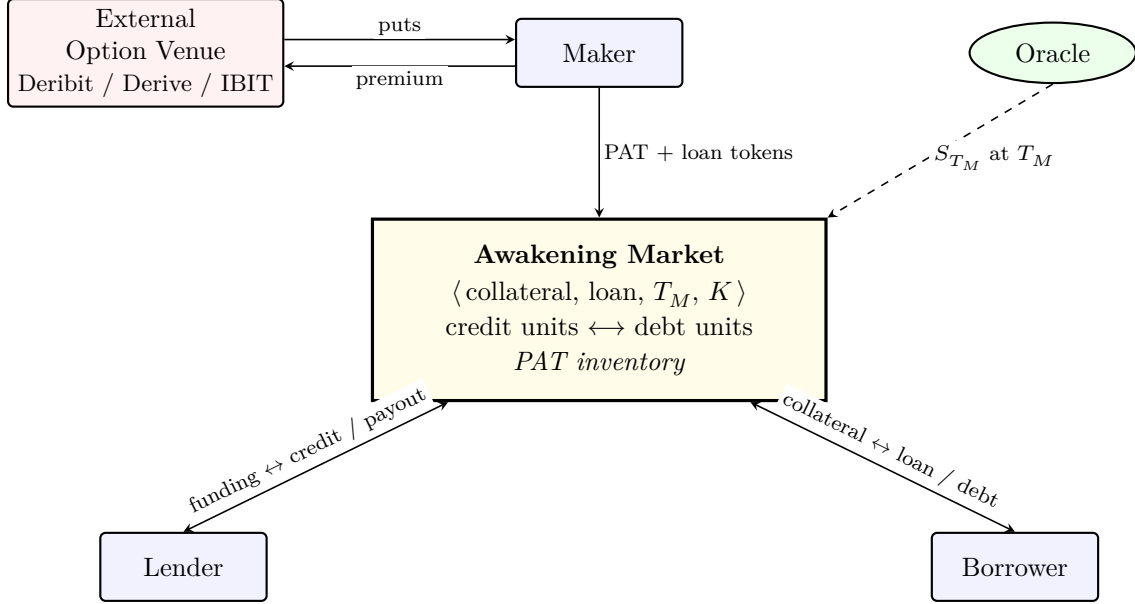


Figure 1: Awakening protocol architecture. The market is the central on-chain contract. Borrowers post collateral and assume debt in exchange for loan tokens. Lenders provide funding and hold fungible credit units. The maker is the conduit between the market and external option venues, sourcing put positions and attributing them to the market via the fill-time callback. The oracle is read only at T_M ; there is no interim oracle path that triggers forced action on borrower positions.

3.2 Maker callbacks

Offers may specify a callback that executes at take time. This lets makers source the loan tokens, and source or attribute the corresponding option position, only when the offer is filled, instead of provisioning them in advance.

The callback discipline is critical to Awakening because the option attribution is the protocol's substitute for liquidation: a maker who has not procured a put position commensurate with the credit they are issuing has, in effect, sold an uncollateralized call to the borrower. Section 4.3 details the constraints enforced on the callback.

In particular, makers can keep the capital backing their offers deployed productively elsewhere until their offer is filled. For example, a maker may keep loan tokens deployed in a Morpho Blue market while quoting a fixed-rate offer on Awakening, and may simultaneously have a standing put position on an external option venue against which Awakening credit can be drawn. If the offer is taken, the callback withdraws the necessary loan tokens, posts the put attribution, and completes settlement within the same transaction.

3.3 Multi-market offers

Callbacks allow makers to publish multiple offers backed by the same liquidity *and the same option inventory*, which is key against fragmentation across maturities and strikes. To handle the liquidity available across different offers, offers can be assigned to a common consumption group that shares a fill budget. Executing any offer in the group decreases the remaining budget for all other offers in it; once the budget is exhausted, no further offer in the group can be filled.

This is particularly useful for makers running an aggregated option book: a maker holding N puts at strike K expiring at T_M can quote against multiple Awakening markets that share these parameters (e.g., different collateral wrappers of the same underlying), with the consumption group bounding total exposure to N .

To make this efficient at scale, ratifiers can support the ratification of a Merkle root of an offer set, allowing makers to quote multiple offers across many markets with a single signature or interaction. These offers can be taken later by presenting the corresponding Merkle proof.

3.4 Routing

The offer publication layer does not guarantee anything about the executability of an offer on Awakening. A taker seeking the best available liquidity across outstanding offers faces a genuine search problem, required to take into account callback execution, consumption groups, gas, *and the maker's residual option inventory after recent fills*. This process, referred to as routing, takes place off protocol and can be performed by anyone.

This makes Awakening fundamentally different from a central limit order book logic: the protocol maintains no canonical queue of resting orders, no protocol-level price-time priority, and no protocol-level reservation of capital.

3.5 Tick structure

While the protocol does not impose a queue for offers, routers will naturally compare them in price order. Without a minimum increment, makers could undercut one another by economically insignificant amounts and the incentive to provide meaningful size would deteriorate.

To prevent this, Awakening enforces a discrete price tick grid for offers, defined identically to Midnight's: consecutive ticks correspond to a constant relative change in implied total return. Let δ denote the target relative increment in implied return, and let N denote the tick range. The unquantized price of tick n is

$$\tilde{P}(n) = \frac{1}{1 + (1 + \delta)^{N/2-n}}$$

Each market starts with ticks corresponding to $\delta = 0.02$ (a 2% relative change in implied return per tick), and may be tightened to $\delta = 0.01$ or $\delta = 0.005$ via a *tickSpacingSetter* role, subject to the invariant that previously valid ticks remain valid.

4 Option attribution

4.1 The non-liquidative invariant

The defining invariant of the protocol is: at maturity, each credit unit in a market M pays one *loan* token, regardless of the terminal collateral price S_{T_M} . This is the *non-liquidative invariant*. It holds without recourse to liquidation, margin call, or oracle-triggered position closure during the term of any loan.

The invariant is sustained by attribution of put option positions to credit units. For a market with strike K , maturity T_M , collateral C , and loan asset L , the attached put has terminal payoff (per unit of collateral notional)

$$\Pi(S_{T_M}) = \max(K - S_{T_M}, 0)$$

denominated in L . Combined with the collateral itself, the lender's per-unit recovery at T_M is

$$R(S_{T_M}) = \min(S_{T_M}, K) + \Pi(S_{T_M}) = K$$

which is exactly the maxDebt of the loan. This identity, evaluated at any terminal price, is the algebraic source of the invariant.

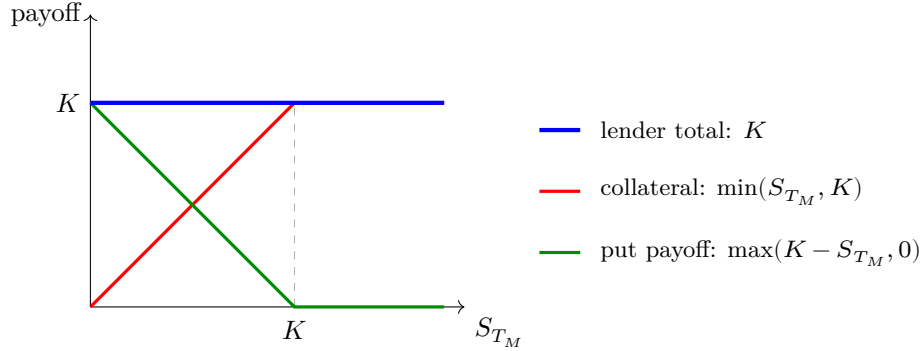


Figure 2: Lender payoff at maturity, per unit of collateral. Collateral value $\min(S_{T_M}, K)$ plus put payoff $\max(K - S_{T_M}, 0)$ sum to the constant K at every terminal price.

4.2 Attribution at fill time

When an offer is filled, the maker callback (Section 3.2) is responsible for delivering both:

1. The loan tokens corresponding to the credit being created, and
2. A put attribution token (PAT) representing a claim on the put payoff $\Pi(S_{T_M})$ scaled to the loan's notional.

The protocol enforces that the PAT supplied to the market satisfies:

- *Underlying alignment*: the PAT references the same collateral asset C as the market.
- *Strike alignment*: the PAT references the same strike K as the market.
- *Maturity alignment*: the PAT references the same maturity T_M as the market, or an expiry within a configurable tolerance window $[T_M - \tau, T_M]$ where τ defaults to zero.
- *Notional sufficiency*: the PAT's underlying notional is at least the collateral notional implied by the loan size.

If any of these constraints fails, the callback reverts, the offer is not filled, and no credit or debt is created.

4.3 PAT sources

A PAT is a thin wrapper around a put position, exposing a uniform settlement interface to the Awakening *settlement* contract. The wrapper is opaque with respect to the put's origination venue. In current practice three classes of source are supported:

1. *On-chain native*: PATs minted against put positions on Derive, Premia, or Panoptic, wrapped by a verification adapter that asserts the underlying option’s parameters at mint time.
2. *Off-chain attested*: PATs minted against put positions held with a qualified custodian (e.g., on Deribit by Coinbase, on IBIT-listed contracts via a regulated U.S. broker), where the custodian attests to the position via a signed feed. Settlement at T_M is funded by the custodian against the attestation.
3. *Protocol-internal*: PATs minted against an Awakening-internal put writer pool, where a separate set of makers post collateral and write puts directly to the protocol. This source exists for markets in which external venues do not list the required strike or maturity.

The PAT abstraction allows a single market to be backed by a heterogeneous mixture of these sources. The market does not need to know which source backs which credit unit; the only invariant is that the aggregate PAT inventory in the market covers the aggregate credit issuance at T_M .

4.4 Premium funding

The cost of the put is borne by the borrower, embedded in the rate at which they sell debt units. For a market with strike K , maturity T_M , and time-to-maturity $\tau = T_M - t$, the borrower’s all-in rate decomposes notionally as

$$r(t) = r_f(\tau) + \frac{P(K, \tau)}{\tau} + m$$

where:

- $r_f(\tau)$ is the term-matched risk-free rate (in practice, a stablecoin term curve).
- $P(K, \tau)$ is the prevailing put premium per unit notional.
- m is the maker’s margin.

The protocol does not enforce this decomposition. It is supplied here as the economic reference that explains why fixed-rate offers in Awakening price the way they do, and why a market’s offer curve will tend to shift in lockstep with the put’s implied volatility on external venues.

4.5 Maker hedging discipline

A maker quoting in Awakening is, in effect, simultaneously:

- Selling a zero-coupon obligation (the credit unit), and
- Holding (or sourcing) a put position to attribute at fill time.

The maker’s net delta with respect to S_{T_M} is, by construction, zero with respect to the credit unit alone (zero-coupon payoff does not depend on S) and matches the put’s delta with respect to the attribution. A maker who holds an opposing put position on the same underlying (sourced from another market-maker on Deribit, for instance) is therefore approximately delta-flat across the combined book.

The protocol does not impose a hedging requirement on makers beyond the per-fill PAT delivery. Makers may run net long or net short option books between fills, subject to their own risk management. The PAT attribution at fill time is sufficient to guarantee the lender’s payoff at T_M .

5 Maturity settlement

5.1 Settlement procedure

At T_M , the market's *settlement* contract executes the following sequence, atomically per market:

1. Read S_{T_M} from the market's oracle.
2. Compute the total credit outstanding C_{T_M} in *loan* tokens.
3. Compute the total collateral outstanding X_{T_M} in *collateral* tokens.
4. Compute the put payoff per collateral unit $\Pi(S_{T_M}) = \max(K - S_{T_M}, 0)$.
5. Aggregate PAT settlement: redeem all PATs in the market, receiving $X_{T_M} \cdot \Pi(S_{T_M})$ in loan tokens from the PAT sources.
6. Open the *repayment window* (Section 5.2).

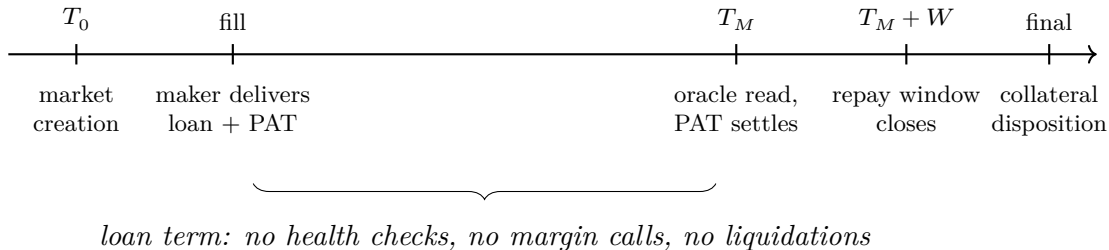


Figure 3: Settlement timeline. Between T_0 and T_M the protocol contains no oracle path that triggers forced action on borrower positions. All settlement is concentrated at and after T_M .

5.2 Repayment window

For a configurable window W (default 24 hours) after T_M , borrowers may repay their outstanding debt L in loan tokens and withdraw their collateral. The repayment amount is exactly the debt; there is no late penalty inside the window.

This is the rational path for borrowers when $S_{T_M} > K$: they recover collateral worth more than the loan amount.

5.3 Collateral disposition after window

If the borrower has not repaid by $T_M + W$, the collateral is released to the settlement contract. The collateral is then aggregated with the put payoff and used to satisfy lender claims:

- Each credit unit is redeemed for one loan token.
- The settlement pool's solvency is the algebraic identity of Section 4.1: collateral value $\min(S, K)$ plus put payoff $\max(K - S, 0)$ equals K per unit of collateral, which is by construction the maxDebt and therefore covers all outstanding credit.

Any residual surplus (which can arise if the put payoff exceeds the credit issuance, due to PAT over-attribution) accrues to the market's *surplus role*, configurable at market creation.

5.4 No interim liquidations

There is no oracle path during $t < T_M$ that triggers any forced action on borrower positions. Specifically:

- No health check during the term.
- No price-triggered margin call.
- No third-party liquidator role active before T_M .

This is the structural difference from Midnight, Morpho Blue, Aave, and Compound, all of which contain liquidation paths during the loan term. The cost of removing those paths is the put premium, which is priced into r at origination.

5.5 Bad debt

In the absence of PAT failure (Section 5.6), there is no bad debt: the non-liquidative invariant guarantees full credit redemption.

5.6 PAT failure

If at T_M a PAT source fails to deliver the contracted put payoff (e.g., a custodial attestor defaults, or an on-chain protocol becomes insolvent), the affected market enters a *shortfall* state. In shortfall:

- The settlement contract redistributes available recoveries pro-rata across credit units.
- Trading in the market remains open post-maturity to allow informed exit.
- Future markets created against the same PAT source may be gated against (Section 7).

PAT source diversification across on-chain native, off-chain attested, and protocol-internal sources (Section 4.3) is the protocol's primary defense against this risk. Source concentration is not enforced at the protocol layer; it is a curatorial decision at market creation and at maker-quoting time.

6 Worked examples

The following examples illustrate the protocol's mechanics with concrete numbers. All figures are illustrative and do not reflect any actual quote. Throughout, the collateral asset is BTC, the loan asset is USDC, and prices are quoted in USDC per BTC. Premium estimates assume a 60% annualized BTC implied volatility on the reference put.

6.1 Example A: at-the-money 90-day loan

At T_0 , BTC trades at $S_0 = \$100,000$. A market M_A is created with $K = \$100,000$ and $T_M = T_0 + 90$ days. The corresponding 90-day, \$100,000-strike BTC put trades on a reference venue at approximately \$5,000 per BTC.

A maker publishes an offer to purchase debt units at price $P = 0.935$, implying an annualized rate of approximately 28%. The borrower posts 1 BTC of collateral and consumes the full offer for a notional of 100,000 debt units.

The maker callback delivers:

- 93,500 USDC to the borrower,
- a PAT representing a \$100,000-strike put on 1 BTC expiring at T_M , sourced from the maker's external put inventory.

The market state after fill: 100,000 credit units issued, 100,000 debt units owed, 1 BTC of collateral locked, 1 BTC of put notional attributed.

Settlement at T_M proceeds along three representative paths:

Case	S_{T_M}	Borrower action	Lender receives	Borrower terminal P&L
1	\$120,000	repays 100,000 USDC	100,000 USDC	+\$13,500
2	\$100,000	indifferent	100,000 USDC	−\$6,500
3	\$60,000	walks away	60,000 + 40,000 (put) = 100,000	−\$6,500

In Case 3, the unhedged BTC holder would have realized a \$40,000 loss against T_0 . The Awakening borrower’s loss is bounded at the loan discount (\$6,500), which is the explicit cost of the embedded put protection. The borrower retains the option to repay and recover the collateral in any state; doing so is rational whenever $S_{T_M} > L = \$100,000$.

6.2 Example B: below-spot strike, lower premium

At T_0 , BTC again trades at $S_0 = \$100,000$. A market M_B is created with $K = \$80,000$ (an 80% advance rate) and $T_M = T_0 + 90$ days. The corresponding 80,000-strike, 90-day put trades at approximately \$2,500 per BTC, materially below the at-the-money premium of Example A.

A maker quotes at $P = 0.965$ (annualized rate $\approx 14\%$). The borrower posts 1 BTC and consumes 80,000 debt units. The callback delivers 77,200 USDC and an 80,000-strike PAT.

Case	S_{T_M}	Borrower action	Lender receives	Borrower terminal P&L
1	\$120,000	repays 80,000	80,000 USDC	+\$37,200
2	\$90,000	repays 80,000	80,000 USDC	+\$7,200
3	\$50,000	walks away	50,000 + 30,000 (put) = 80,000	−\$2,800

Relative to Example A, the borrower trades a lower advance (\$77,200 vs \$93,500) for a substantially lower funding cost and a wider downside before the put begins to compensate.

6.3 Example C: comparison to a Midnight-style liquidation loan

Consider a hypothetical Midnight market on the same collateral with a liquidation LTV of 80% and a fixed annualized rate of $r_M = 10\%$. Compare to Example B:

- The Midnight loan funds at a lower rate (10% vs 14%). The 4% spread is, mechanically, the implied premium for path-independence.
- During the term, if S_t falls below the liquidation threshold even transiently, the Midnight position is liquidatable. The borrower forfeits collateral at the liquidation price, regardless of S_{T_M} .
- The Awakening position is not liquidatable during the term. Consider a path with $S_{T_M/2} = \$50,000$ (deep below the liquidation threshold) but $S_{T_M} = \$90,000$:
 - The Midnight borrower has been liquidated and has no collateral to recover.
 - The Awakening borrower repays \$80,000 at T_M and retrieves BTC worth \$90,000.

The path-dependence premium is the economic value the borrower pays for the option to ride a drawdown. Whether this premium is worth paying is a function of the borrower’s view on intra-term volatility and on the conditional distribution of recovery paths.

7 Access-control gates

Awakening is designed to support flexible access-control conditions. Markets may specify up to two optional gate contracts at creation, fixed thereafter, which the protocol calls when gated actions are attempted. This lets each market embed conditions specific to its intended use.

The *enter gate* restricts who may increase their credit or debt position. It applies only to entry: a participant can always exit by withdrawing credit, repaying debt, or recovering collateral, even if the gate reverts. Because the gate is an external contract that may evolve over time, this restriction ensures that it cannot trap funds inside a market and remains an access-control layer rather than a custody risk.

The *PAT gate* restricts which PAT sources are admissible for the market. This is a market-creation parameter and allows curators to limit a market's option attribution to, for instance, only on-chain native sources, or only sources with a minimum credit rating from a designated attester.

8 Authorizations

Awakening uses a single coarse-grained authorization primitive. An account may authorize another address to act on its behalf. Authorization is deliberately not scoped by action or by market: once granted, it gives the authorized address full control over the authorizer's Awakening state, including the ability to add or remove other authorizations.

This choice gives a lot of flexibility, but comes with a trade-off: scoped delegation is not expressible natively. Integrations that want to grant a third party only a subset of actions must do so through an intermediate contract that holds the authorization and exposes a narrower interface to its own users. Granularity therefore lives on top of the protocol rather than inside it.

9 Fees

Awakening can charge two protocol-level fees on each market: a settlement fee on each trade and a continuous fee that accrues over time on outstanding credit. Both are bounded by immutable caps, giving participants a permanent upper bound on the fees the protocol can charge. Default fee values are configured per loan token and can be overridden per market. A designated role sets these values, while another role receives the accrued fees.

9.1 Settlement fee

The settlement fee is time-to-maturity dependent and is charged at take time. The fee inserts a spread between the buyer's and seller's settlement prices. Given a buy offer (resp. sell offer) at price P , the taker sells at $P - f(\text{TTM})$ (resp. buys at $P + f(\text{TTM})$).

The fee rate $f(\text{TTM})$ is a piecewise linear continuous function in time to maturity, with breakpoints at 0, 1, 7, 30, 90, 180 and 360 days, and constant after maturity and above 360 days. The fee value at each breakpoint is bounded by a hardcoded maximum. The maxima are calibrated so that the implied annualized rate $f(\text{TTM}) \cdot 360/\text{TTM}$ does not exceed 50 bps.

9.2 Continuous fee

The continuous fee accrues each second on outstanding credit and is borne by lenders. It materializes when the lender interacts with the market with a reduction in their credit. Lenders crystalize their future fees when increasing their credit; therefore, later changes in the continuous fee value do not impact existing lending positions. The continuous fee is capped at 1% annualized.

9.3 PAT attribution fee

A third, Awakening-specific fee is charged on each PAT attribution at fill time, denominated as a fraction of the put premium. This fee compensates the protocol for the option-verification adapters maintained for each PAT source class. The fee is bounded at 5% of the attributed premium and is paid by the maker (not the borrower), in loan tokens.

10 Comparison to existing protocols

10.1 vs Morpho Midnight

Midnight provides the intent-based, fixed-term, zero-coupon primitive on which Awakening builds. The differences are:

- Midnight is liquidation-based; Awakening is option-attribution-based.
- Midnight's loan-to-value is volatility-bounded; Awakening's is strike-bounded and can exceed 100%.
- Midnight's settlement is asynchronous (positions can be liquidated piecewise during the term); Awakening's is synchronous (settlement occurs at T_M).
- Midnight's maker callback sources loan tokens only; Awakening's sources loan tokens plus PAT.

A market in Midnight and a market in Awakening at the same collateral, loan, and maturity, with respective liquidation LTV and Awakening strike chosen such that they imply equivalent advance rates, are economically substitutable for the borrower except for (a) the price (Awakening will charge the put premium) and (b) the elimination of interim liquidation risk.

10.2 vs Morpho Blue / Aave / Compound

These protocols serve highly liquid collateral with continuous, oracle-driven liquidation. Awakening serves the demand for long-duration, drawdown-tolerant exposure. The two designs are complementary: a borrower seeking maximum capital efficiency on liquid collateral with no preference for path-independence will choose Morpho Blue; a borrower with the opposite preference profile will choose Awakening.

10.3 vs Matrixport NL Loan / off-chain bilateral

Off-chain non-liquidative loans (NL Loan, Genesis structured loans, Galaxy bespoke facilities) are bilateral, opaque, custodial, and not composable. Awakening reproduces the same payoff structure on a permissionless, non-custodial substrate, with a transparent fee path and standardized market parameters that enable a competitive maker market. The cost is the protocol's dependence on a maturing PAT source ecosystem; the benefit is the elimination of single-counterparty credit risk on the lender side.

11 Extensions: zero-coupon and collared products

The base Awakening market attaches a single option claim -a put- to each loan, with the cost borne by the borrower through the loan discount. The same offer-based, fixed-term primitive admits a natural generalization: the market parameters can specify *multiple* option claims, with premiums flowing in either direction. This unlocks a family of structured products on the same infrastructure, including the zero-coupon collared loan implemented bilaterally off-chain by Matrixport's Zero Cost Loan and analogous facilities offered by Galaxy, Cumberland, and other prime brokers.

11.1 Generalized market specification

A generalized Awakening market is specified by

$$M = \langle \text{collateral, loan, oracle, } T_M, K_{\text{put}}, K_{\text{call}}, \text{settlement, gates} \rangle$$

with $K_{\text{put}} \leq K_{\text{call}}$. The base product corresponds to $K_{\text{call}} = +\infty$. The settlement contract attributes both a long put (to the credit side) and a short call (to the debt side) at fill time. The maker callback is correspondingly extended to deliver, alongside the loan tokens, a put attribution token at strike K_{put} and a call attribution token at strike K_{call} , with the same alignment constraints as in Section 4.2.

11.2 Borrower and lender payoffs

At T_M , per unit of collateral:

- The lender receives $\min(S_{T_M}, K_{\text{put}}) + \max(K_{\text{put}} - S_{T_M}, 0) = K_{\text{put}}$, preserving the non-liquidative invariant on the credit side.
- The borrower's terminal collateral claim is $\min(S_{T_M}, K_{\text{call}})$, because the attached short call assigns away any upside above K_{call} .

The borrower's effective payoff profile is therefore a *collar*: floored below at K_{put} (the floor implicit in the loan amount) and capped above at K_{call} (the ceiling implicit in the assigned call).

11.3 The zero-coupon case

Let $P(K, \tau)$ denote the put premium at strike K and tenor τ , and $C(K, \tau)$ denote the call premium at strike K . The maker's break-even discount on the borrower's debt unit is approximately

$$1 - P_{\text{offer}} \approx r_f \cdot \tau + P(K_{\text{put}}, \tau) - C(K_{\text{call}}, \tau) + m \cdot \tau$$

where r_f is the term-matched risk-free rate and m is the maker's margin. If K_{call} is chosen such that

$$C(K_{\text{call}}, \tau) = P(K_{\text{put}}, \tau) + r_f \cdot \tau$$

the discount collapses to $m \cdot \tau$. The borrower receives essentially the full advance at T_0 and owes exactly the same amount at T_M : a *zero-coupon loan*. The implied price is the surrender of all collateral upside above K_{call} .

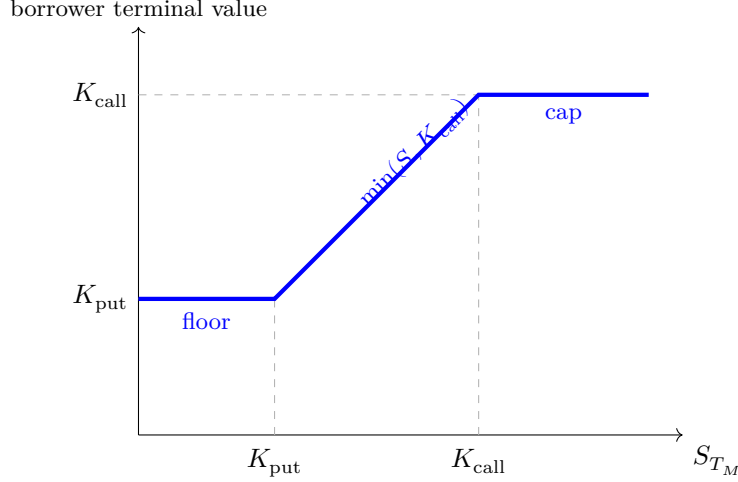


Figure 4: Borrower terminal value under the generalized collared market, per unit of collateral, net of loan repayment. Downside is floored at K_{put} (the loan amount); upside above K_{call} is surrendered to the call holder.

For BTC at $K_{\text{put}} = S_0$, the zero-coupon call strike depends on the prevailing volatility surface but is empirically in the range $K_{\text{call}} \in [1.15 \cdot S_0, 1.30 \cdot S_0]$ for $\tau \in [60, 180]$ days at the implied volatilities observed during 2025-2026. Tighter call strikes are achievable when the put strike is set below spot, since the financed premium is smaller.

11.4 Other payoff profiles

The same generalization expresses other points on the borrower's risk-payoff frontier:

- **Yield-enhanced loan** (K_{call} tight relative to S_0 , $K_{\text{put}} = S_0$): the borrower receives *more* than the loan amount at T_0 (a negative effective rate) in exchange for a tight upside cap. Suitable for borrowers with a high-conviction range-bound view.
- **Wide collar** (K_{call} wide, partial premium offset): the borrower pays a positive but reduced rate and retains most upside. The intermediate point on the frontier.
- **Risk reversal loan** ($K_{\text{put}} \ll S_0$, K_{call} modestly above S_0): the borrower accepts a tail of unhedged downside (below K_{put}) in exchange for a tighter premium and a higher cap. The loan amount is bounded by K_{put} , which is below spot.

Each is a distinct point on the same two-dimensional surface parameterized by $(K_{\text{put}}, K_{\text{call}})$. The protocol does not privilege any region of this surface; market creators select the parameters and quoting makers signal their willingness to underwrite the resulting position by publishing offers.

11.5 Maker hedging across the surface

For a maker quoting in a generalized market, the natural hedge on external venues is the corresponding collar: long put at K_{put} , short call at K_{call} . Collar packages are routinely quoted as single-line block trades on Deribit and through Paradigm's RFQ network. The maker's combined position -credit unit short, collar long- has zero terminal S_{T_M} sensitivity between the strikes, exactly the region in which the borrower's position is also flat.

The protocol's contribution is the standardization, fungibility, and composability of the resulting

borrower-side claim, which today exists only as bespoke bilateral paper outside of the largest prime brokers' counterparty networks.

12 Conclusion

Awakening generalizes Morpho Midnight's offer-based, fixed-rate, fixed-maturity primitive to the non-liquidative case. The single architectural change -attribution of a put option to each credit unit at fill time- is sufficient to remove interim liquidation from the lending path while preserving every other property that makes the Midnight primitive viable on-chain: isolated immutable markets, fungible credit and debt units, capital-efficient maker offers, and multi-market quoting.

The protocol's viability depends on a maturing option-attribution ecosystem. As the on-chain options market deepens (Derive, Premia, Panoptic) and as institutional bridges to centralized option venues (Deribit, IBIT) become standardized, the cost of PAT sourcing will compress, and the spread between Awakening rates and Midnight rates will narrow toward the residual put premium. The path-independent product that today exists only in bilateral off-chain financing becomes available on a permissionless substrate, with the fee transparency, composability, and counterparty-risk profile of on-chain credit.

Future work includes:

- *Strike-rolling markets*: pre-committed sequences of markets at $(T_1, K_1), (T_2, K_2), \dots$ that let borrowers obtain effective tenors longer than the longest individually-quoted PAT, by atomic roll-over at each T_i .
- *Cross-margin between markets*: allowing a borrower's collateral to be shared across multiple Awakening markets, recognizing the natural diversification across strikes and tenors.
- *PAT origination by the protocol*: extending the protocol-internal PAT writer pool to operate as a primary venue, displacing dependence on external option venues for markets at non-standard strikes.

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